# **ME2102:** Engineering Innovation and Modelling

## Power Screws

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# Introduction

- What's a power screw?

# *a.k.a.* lead screw or translational screw, a screw used as a **linkage** in a machine, to translate turning motion into linear motion (Dictionary)



# Mechanical device that converts rotational (turning) motion into linear (straight-line) motion (Wikipedia)

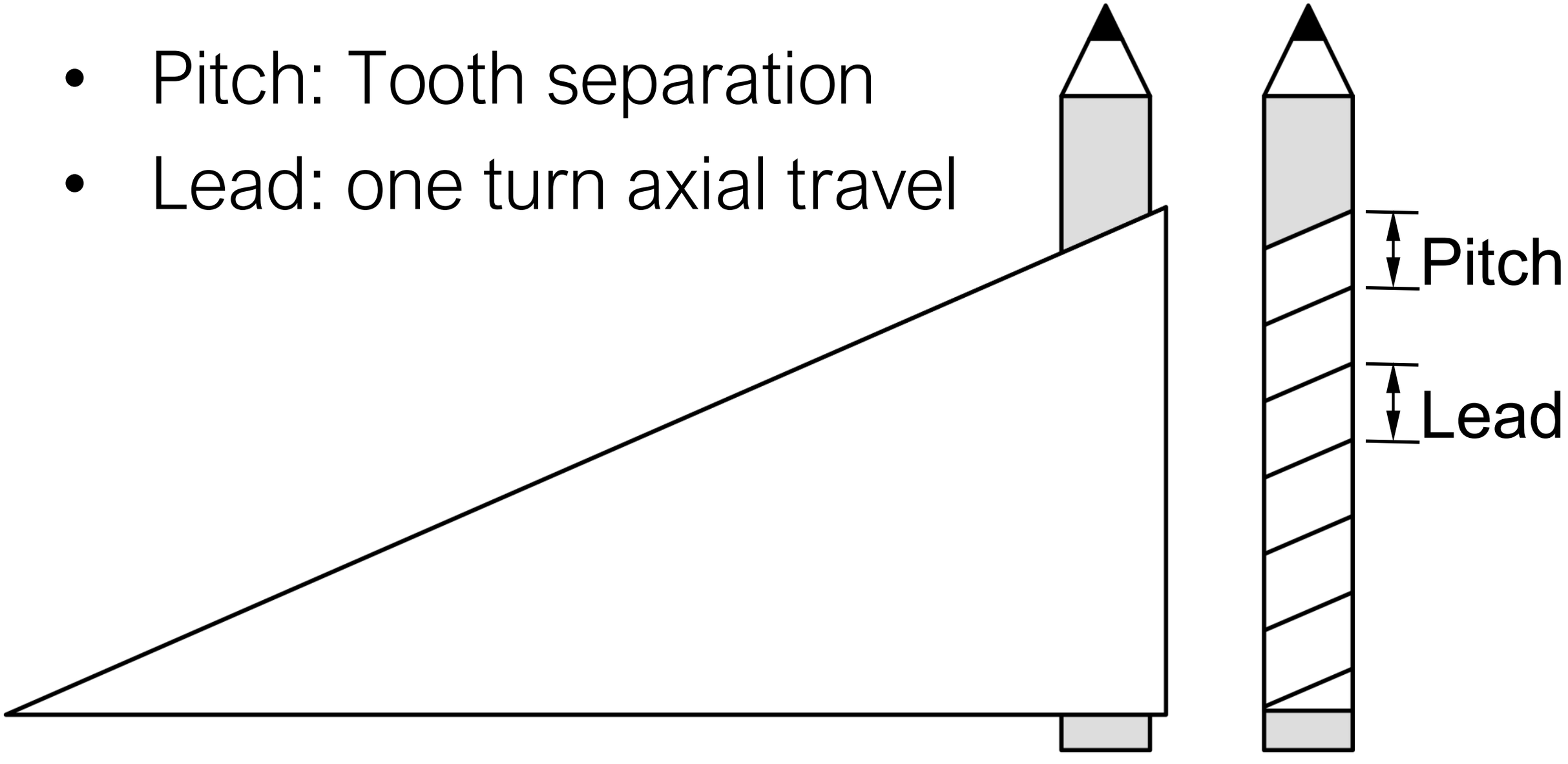
# Mechanical device that converts rotation into linear motion

# Power Screws

- Review screw thread terminology
- Common power screw thread profiles
- Raising and lowering torques
- Collar torque
- Raising efficiency
- Solve problems for power screw design

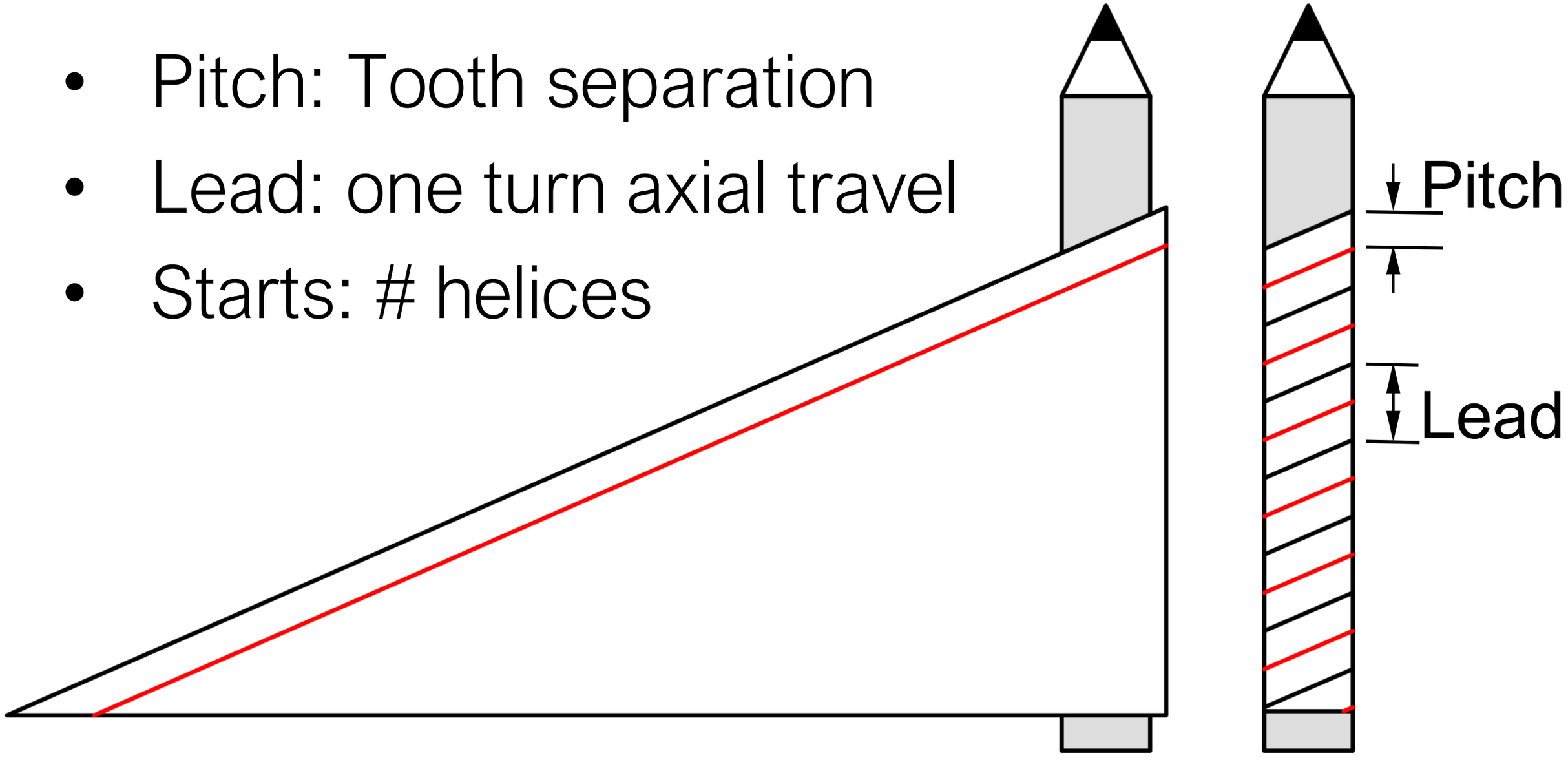
# Screws

- Pitch: Tooth separation
- Lead: one turn axial travel



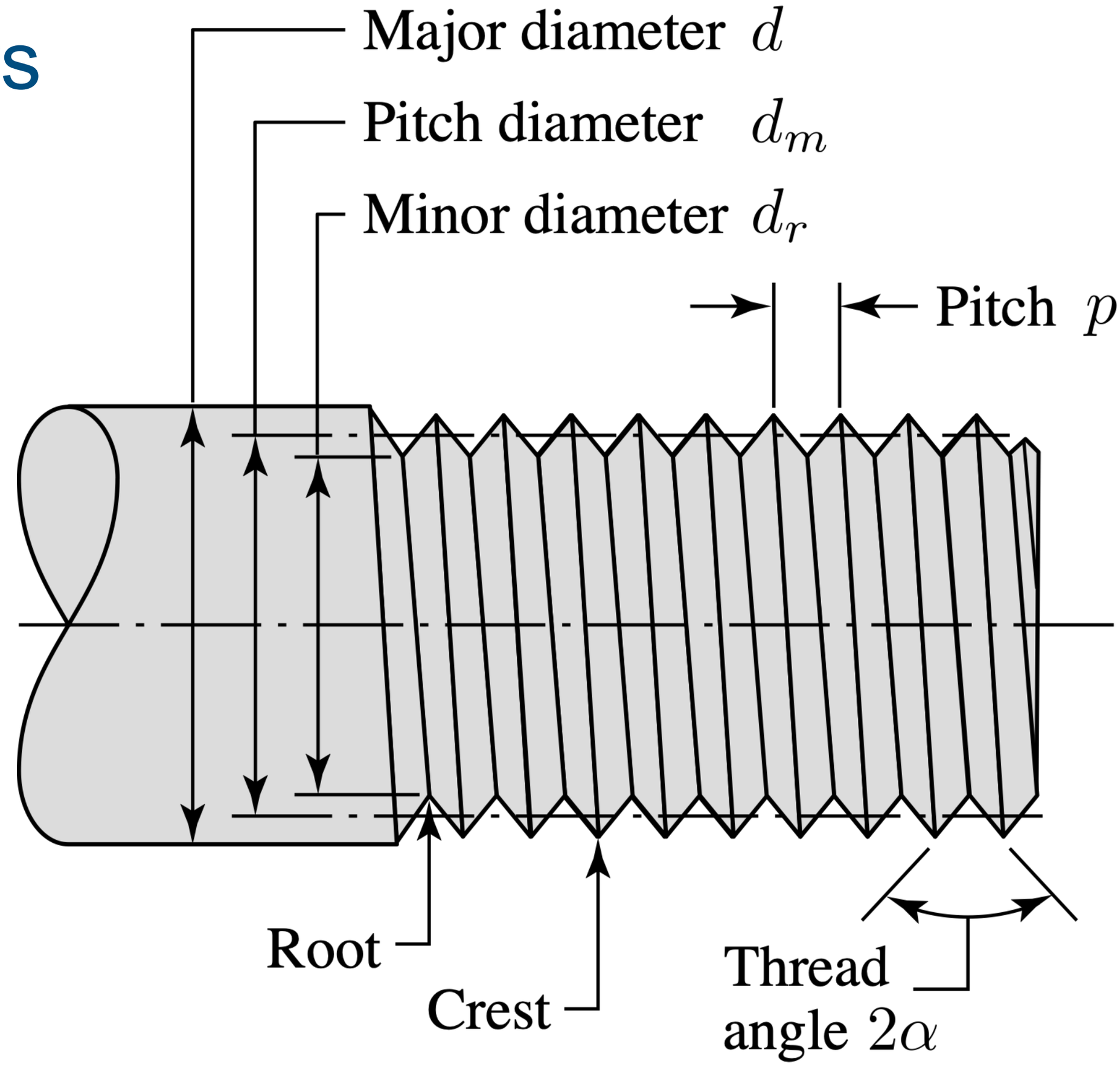
# Screws

- Pitch: Tooth separation
- Lead: one turn axial travel
- Starts: # helices

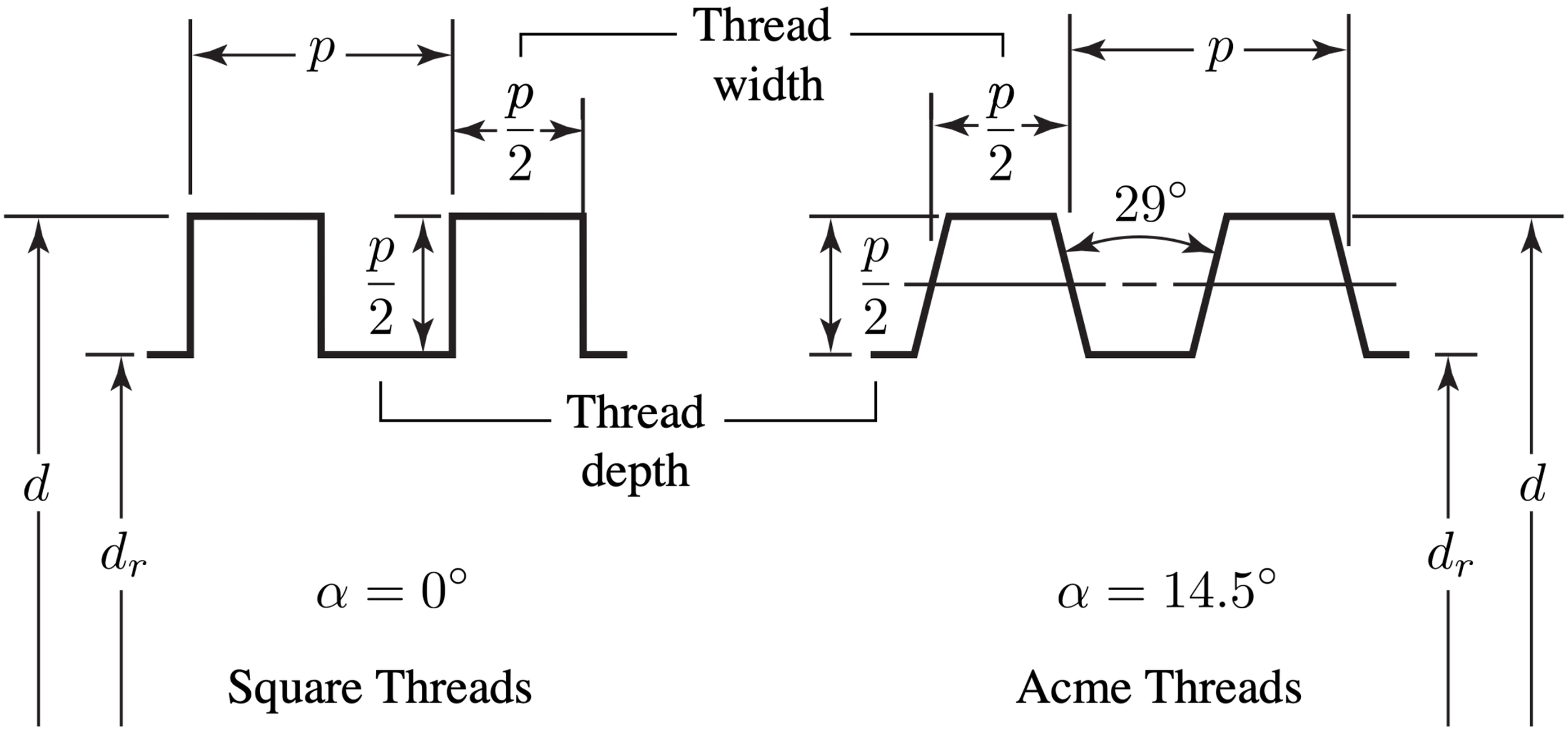



Lead = (Starts) x Pitch

# Screws


Major diameter $d$
Pitch diameter $d_m$
Minor diameter $d_r$
Pitch $p$
Root
Crest
Thread angle $2\alpha$

# Thread Profiles

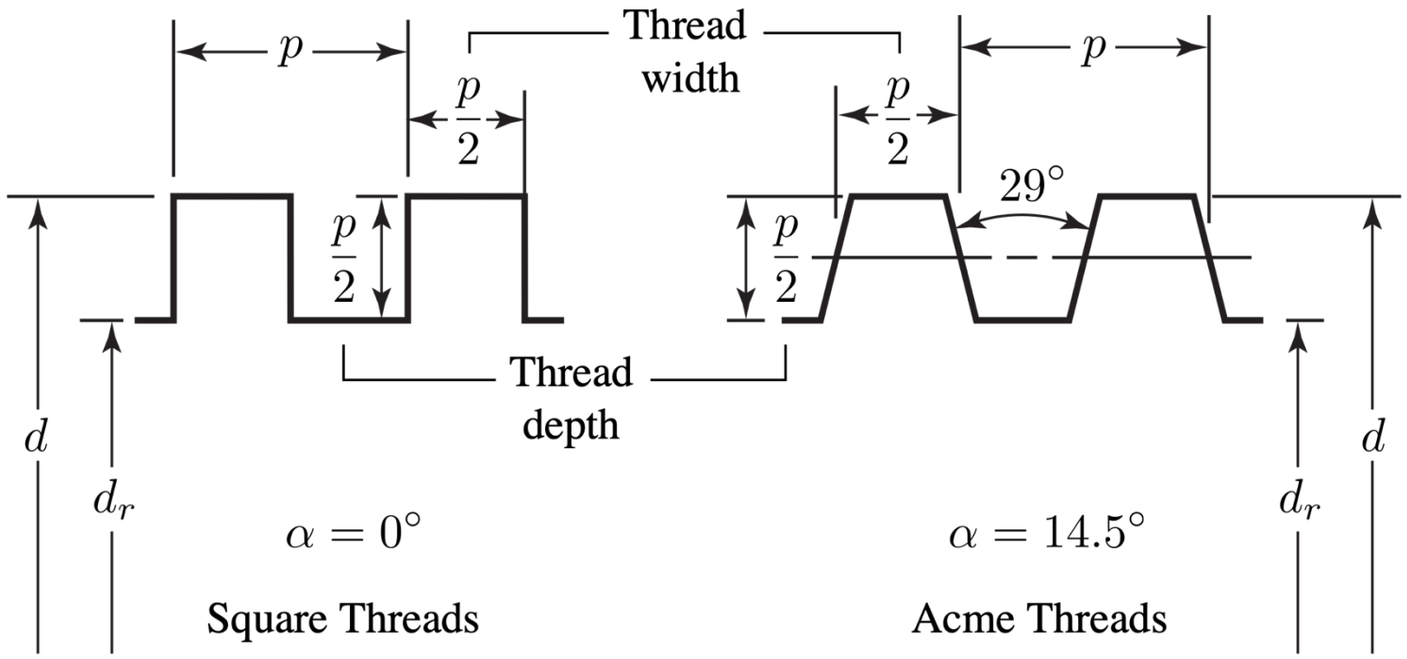



$$d_r = d - 2(p/2) = d - p$$

$$d_m = (d + d_r)/2 = d - p/2$$

# Practice

A square thread screw has diameter 30 mm and pitch 4mm. What is its **thread width**?




$$d_r = d - 2(p/2) = d - p$$

$$d_m = (d + d_r)/2 = d - p/2$$

A) 1 mm

B) 2 mm

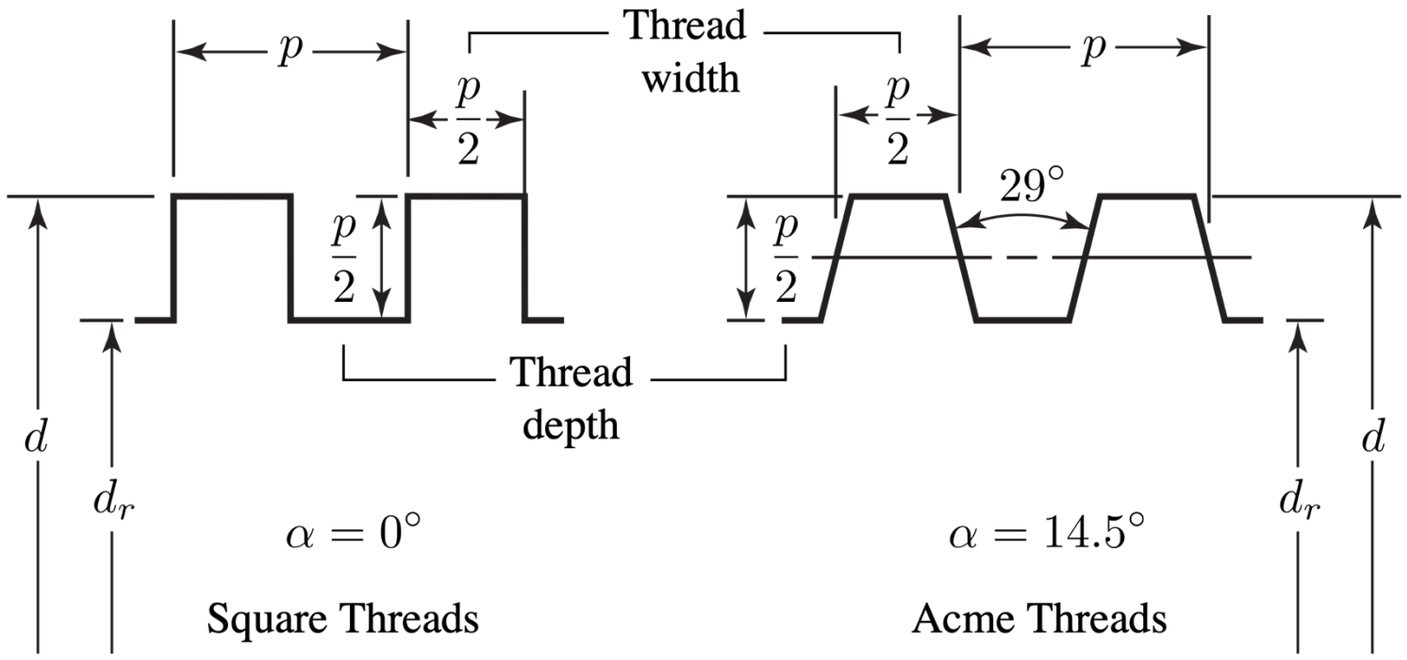
C) 3 mm

D) 4 mm



# Practice

An acme thread screw has diameter 30 mm and pitch 4mm. What is its **thread width**?




$$d_r = d - 2(p/2) = d - p$$

$$d_m = (d + d_r)/2 = d - p/2$$

A 0

B 0

C 0

D 0

A) 1 mm

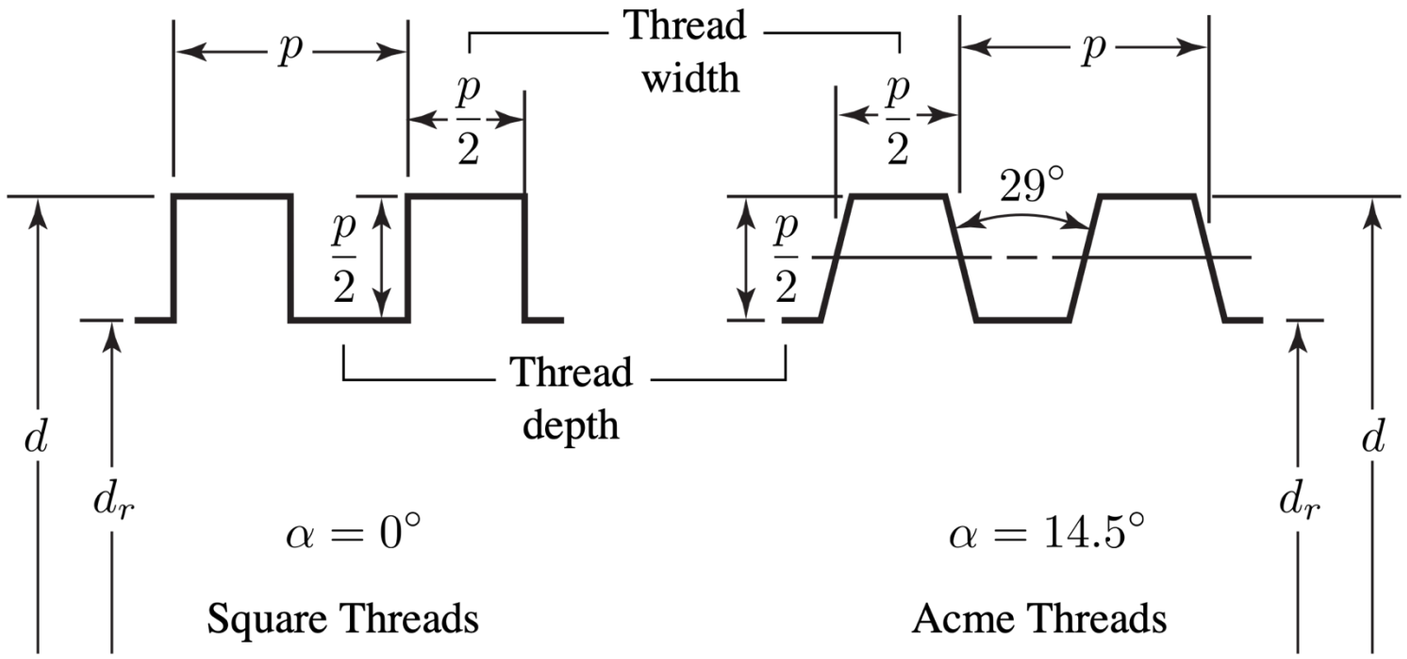
B) 2 mm

C) 3 mm

D) 4 mm

# Practice

A square thread screw has diameter 30 mm and pitch 4mm. What is its **pitch diameter**?




$$d_r = d - 2(p/2) = d - p$$

$$d_m = (d + d_r)/2 = d - p/2$$




A) 32 mm

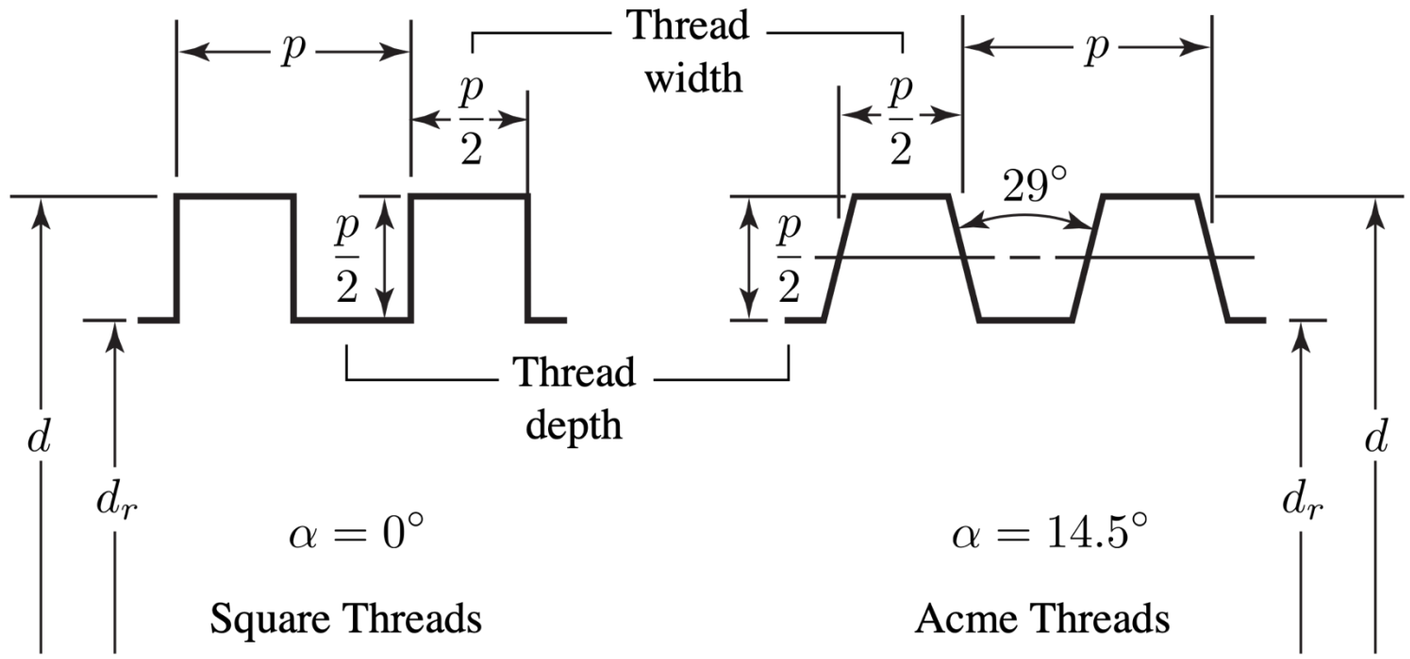
B) 30 mm

C) 28 mm

D) 26 mm

# Practice

An acme thread screw has diameter 30 mm and pitch 4mm. What is its **pitch diameter**?




$$d_r = d - 2(p/2) = d - p$$

$$d_m = (d + d_r)/2 = d - p/2$$

A) 32 mm

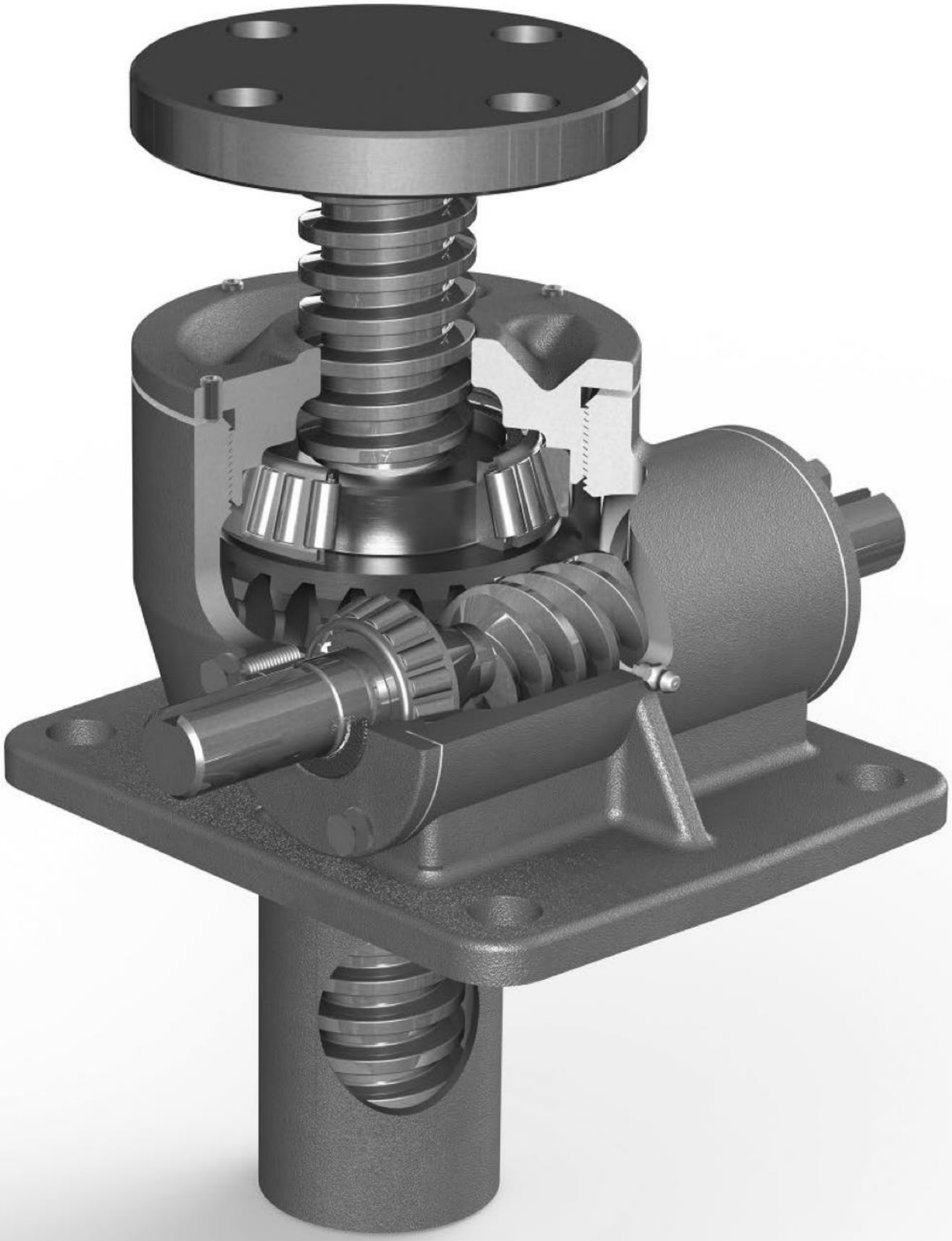
B) 30 mm

C) 28 mm

D) 26 mm

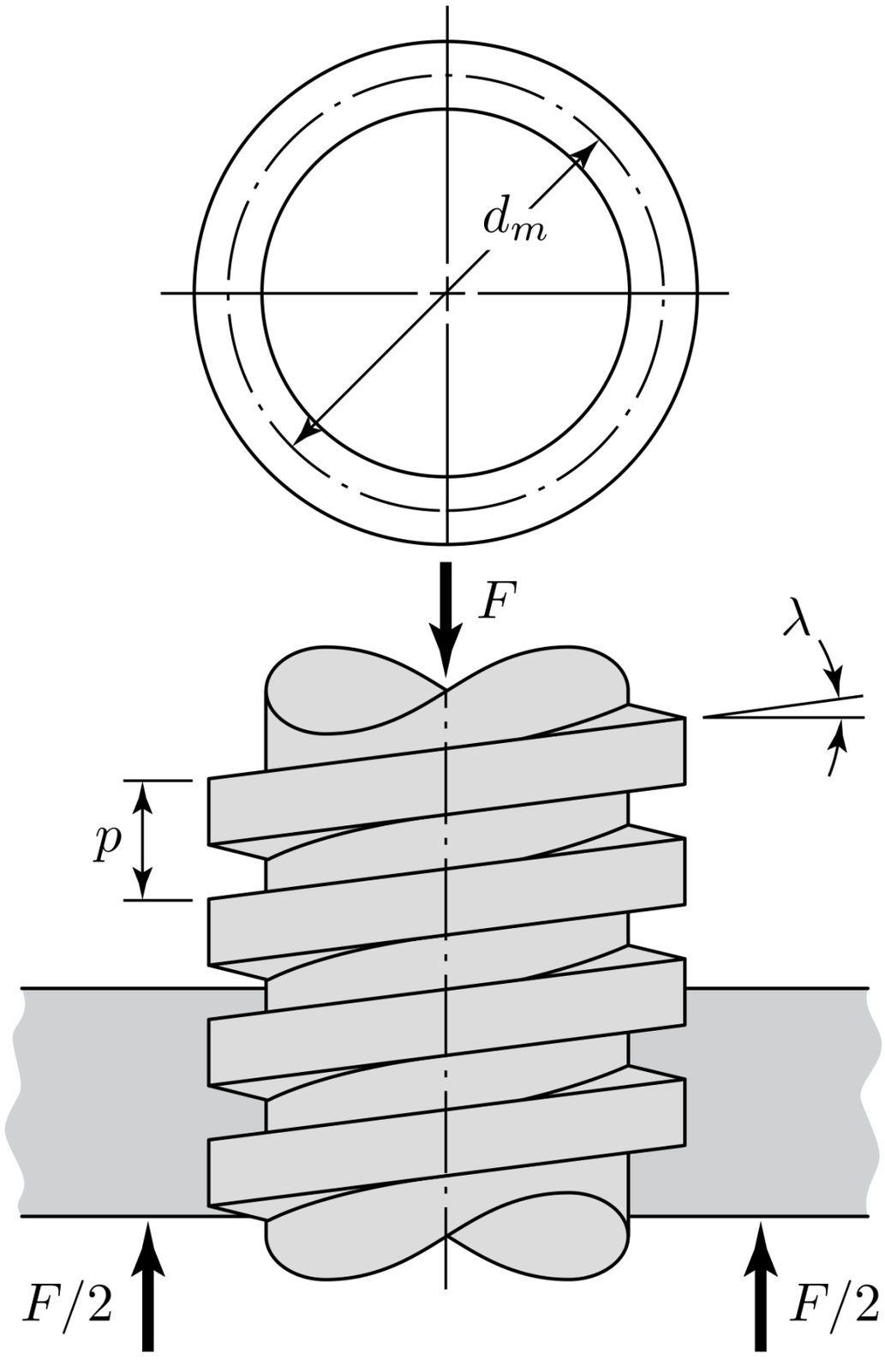


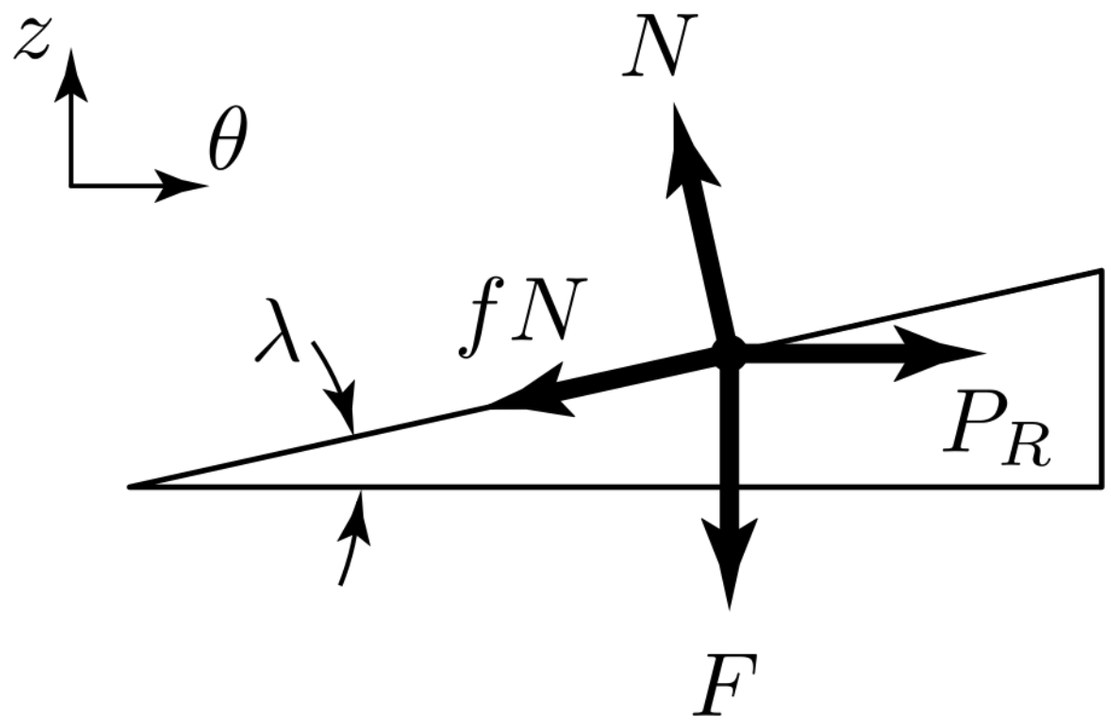
# Power Screws





# Raising Torque



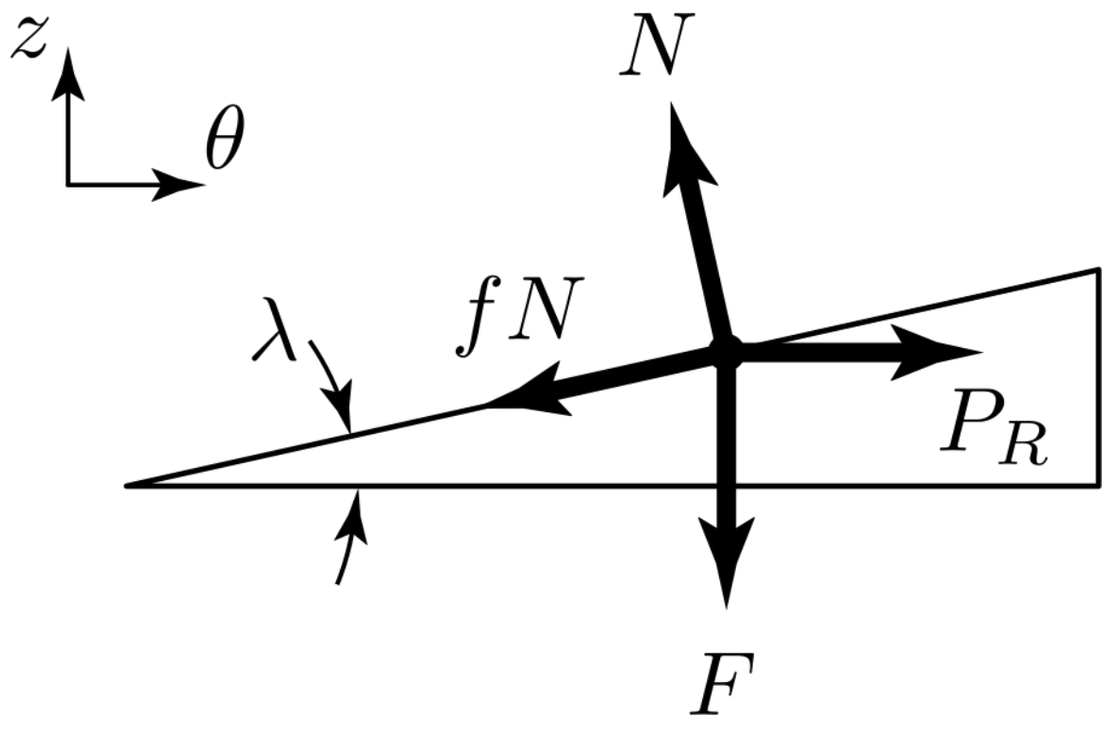

$$T_R = P_R \frac{d_m}{2}$$

$$\tan\lambda = \frac{\ell}{\pi d_m}$$

# Raising Torque

$$\sum F_\theta \quad P_R = fN\cos\lambda + N\sin\lambda$$

$$\sum F_z \quad F = N\cos\lambda - fN\sin\lambda$$




$$\frac{P_R}{F} = \frac{f\cos\lambda + \sin\lambda}{\cos\lambda - f\sin\lambda} = \frac{f + \tan\lambda}{1 - f\tan\lambda}$$

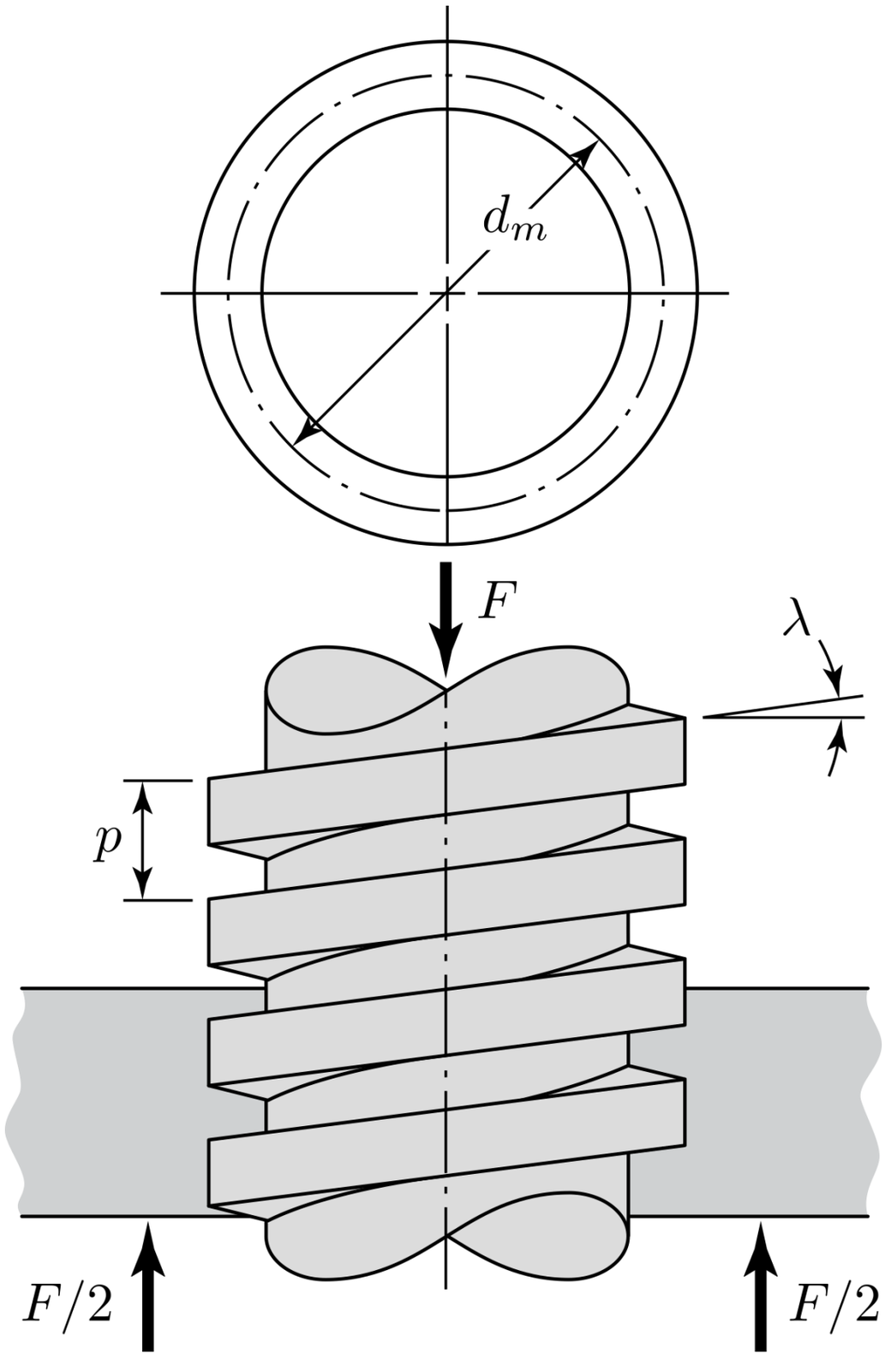
$$T_R = P_R\frac{d_m}{2}$$

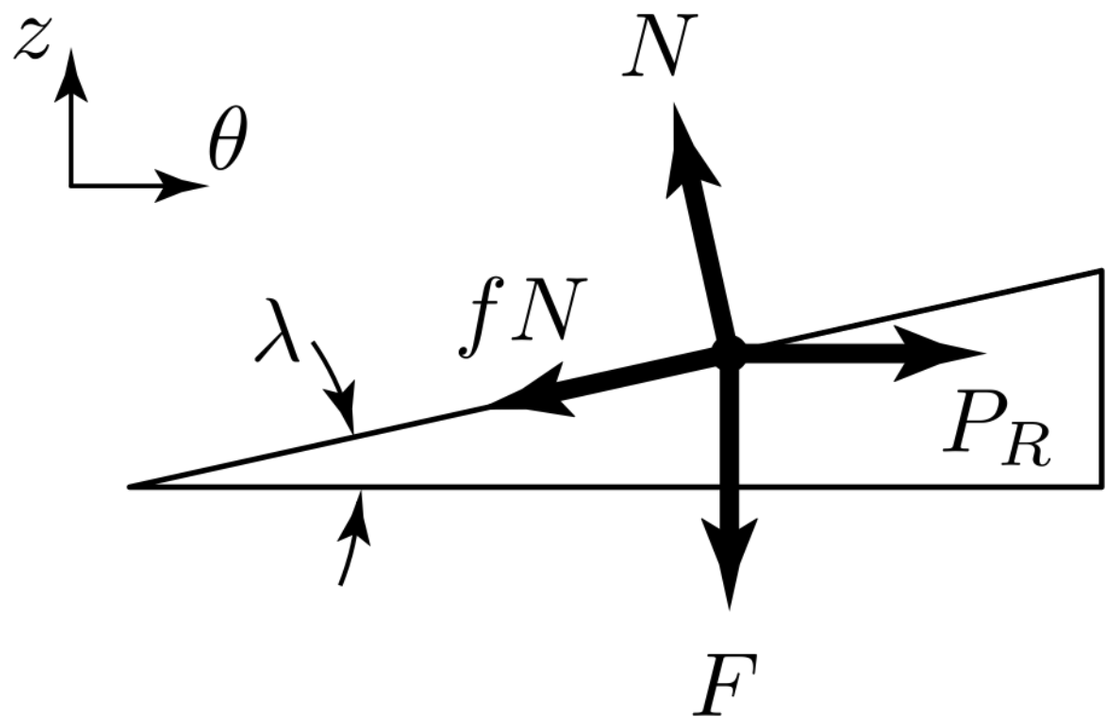
$$T_R = \frac{Fd_m}{2}\left(\frac{\pi d_m f + \ell}{\pi d_m - \ell f}\right)$$

$$\tan\lambda = \frac{\ell}{\pi d_m}$$

larger lead
larger torque
assume $\ell < d_m$

# Raising Torque

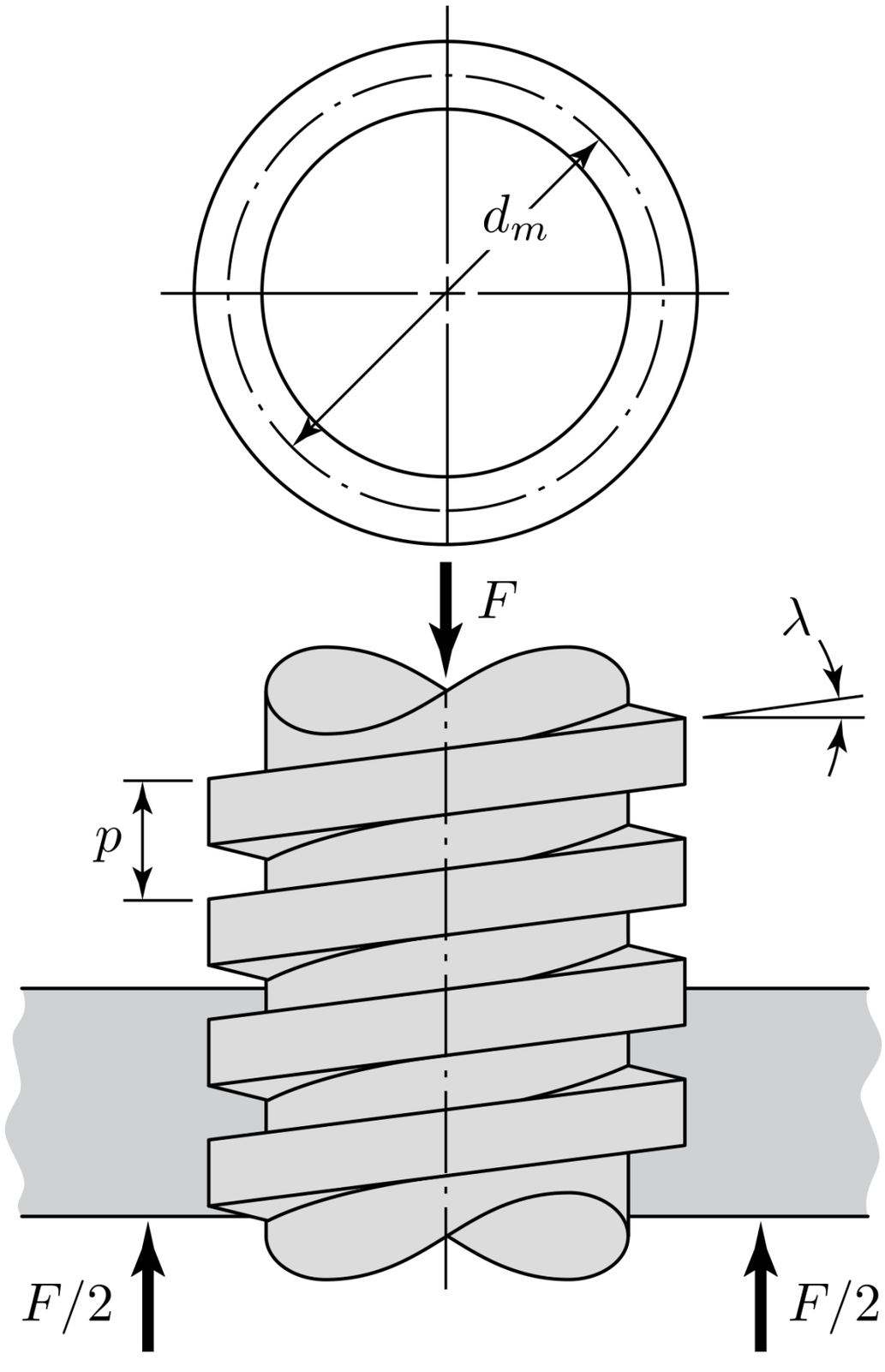


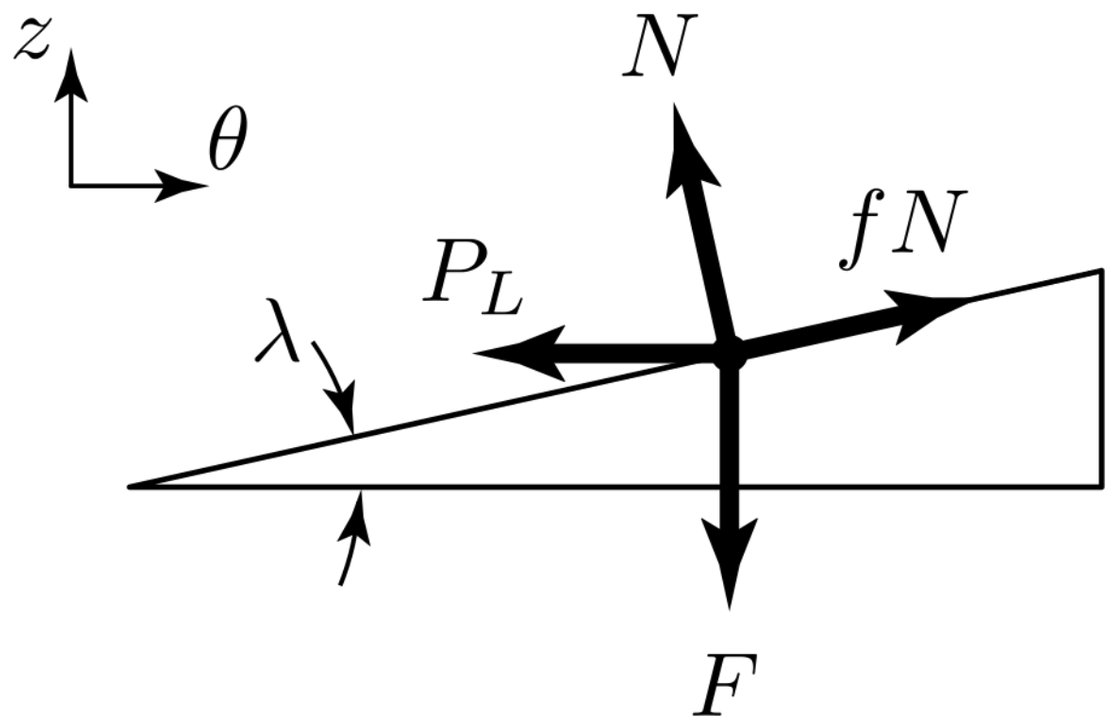




$$T_R = P_R \frac{d_m}{2}$$

$$\tan\lambda = \frac{\ell}{\pi d_m}$$

# Lowering Torque



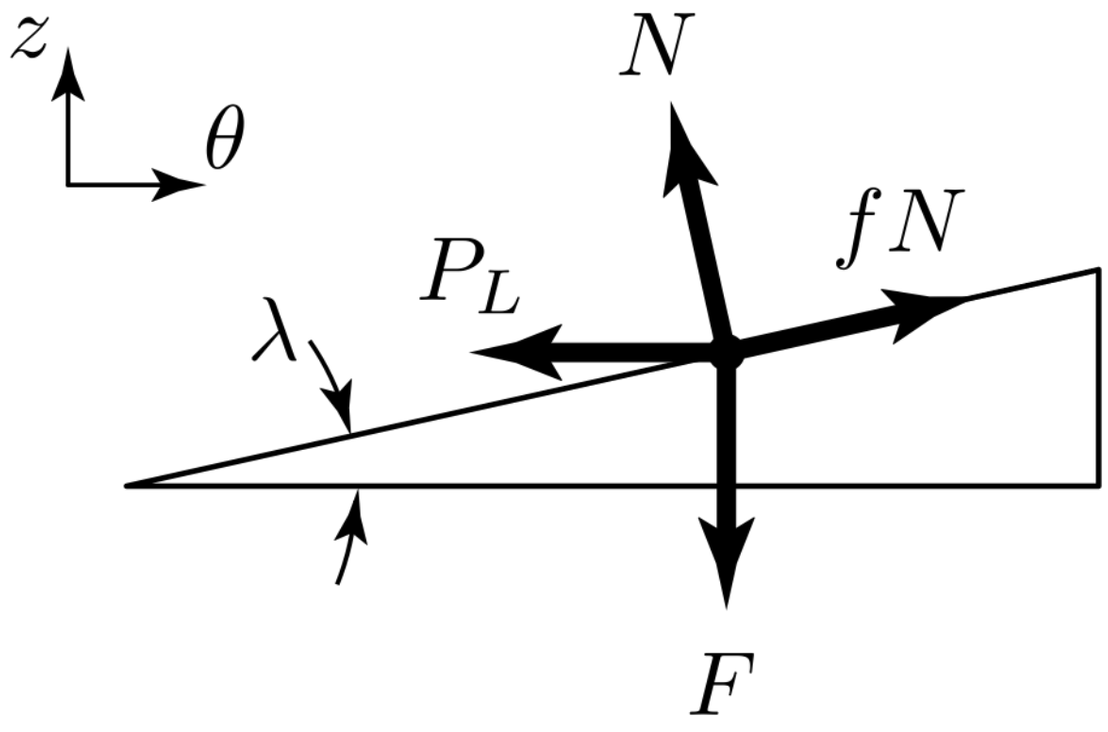

$$T_L = P_L \frac{d_m}{2}$$

$$\tan\lambda = \frac{\ell}{\pi d_m}$$

# Lowering Torque

$$\sum F_\theta \quad P_L = f N \cos\lambda - N \sin\lambda$$

$$\sum F_z \quad F = N \cos\lambda + f N \sin\lambda$$




$$\frac{P_L}{F} = \frac{f\cos\lambda - \sin\lambda}{\cos\lambda + f\sin\lambda} = \frac{f - \tan\lambda}{1 + f\tan\lambda}$$

$$T_L = P_L \frac{d_m}{2}$$

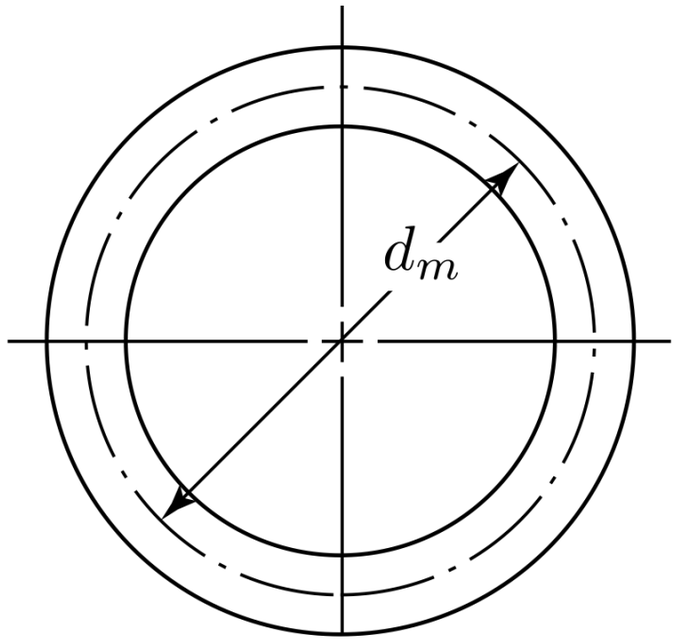
$$T_L = \frac{F d_m}{2}\left(\frac{\pi d_m f - \ell}{\pi d_m + \ell f}\right)$$

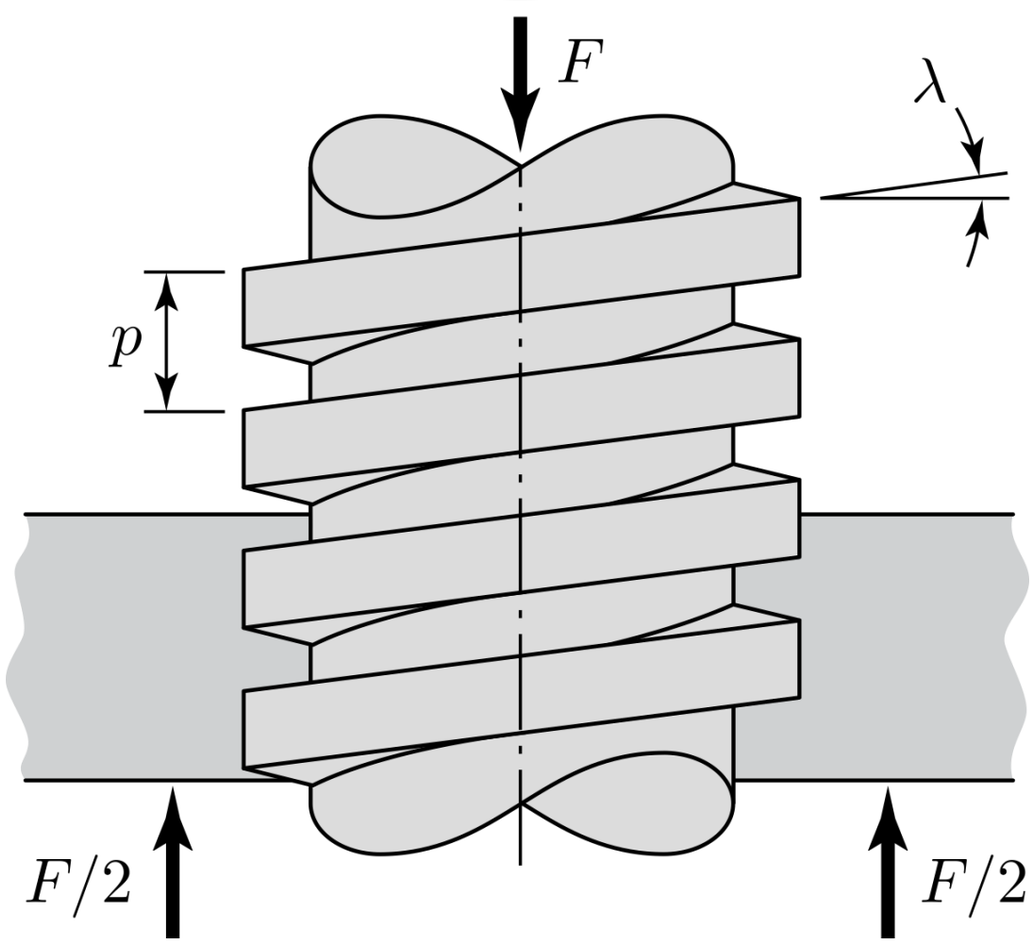
$$\tan\lambda = \frac{\ell}{\pi d_m}$$

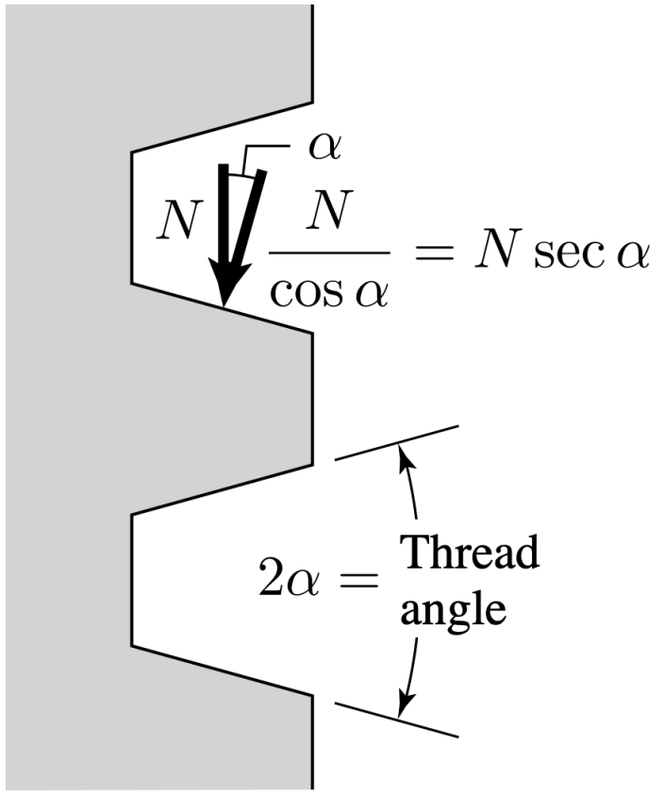
< 0 sliding

> 0 self-locking

# Power Screw Torque



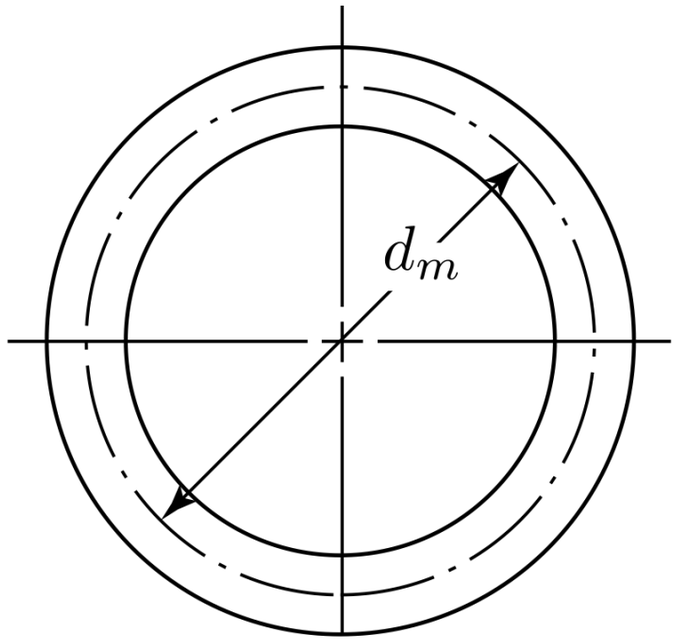

## Raising

$$T_R = \frac{F d_m}{2}\left(\frac{\pi d_m f + \ell}{\pi d_m - \ell f}\right)$$

## Lowering

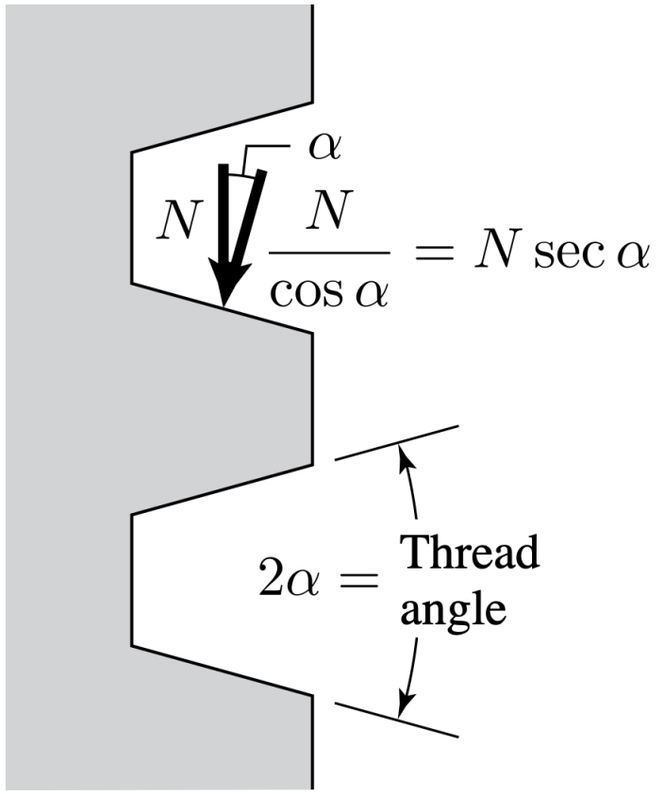
$$T_L = \frac{F d_m}{2}\left(\frac{\pi d_m f - \ell}{\pi d_m + \ell f}\right)$$

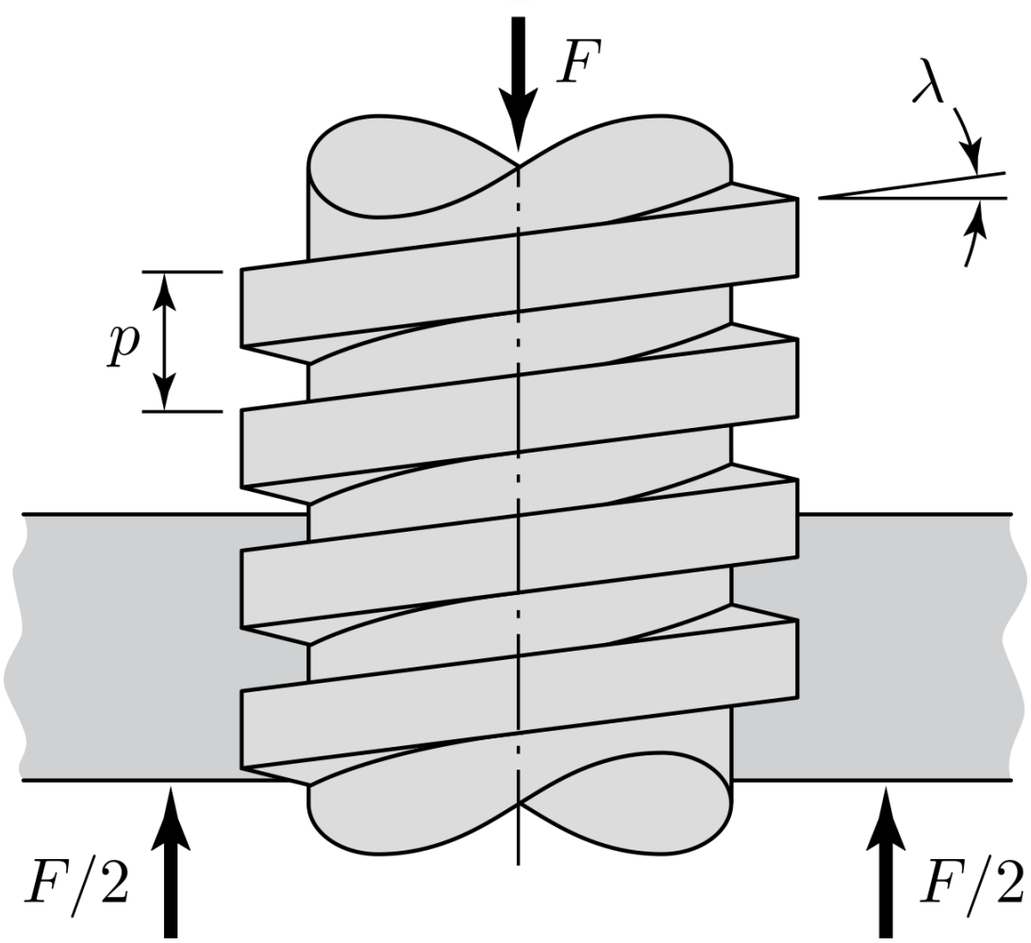
# Power Screw Torque




Square $\alpha = 0°$

Acme $\alpha = 14.5°$



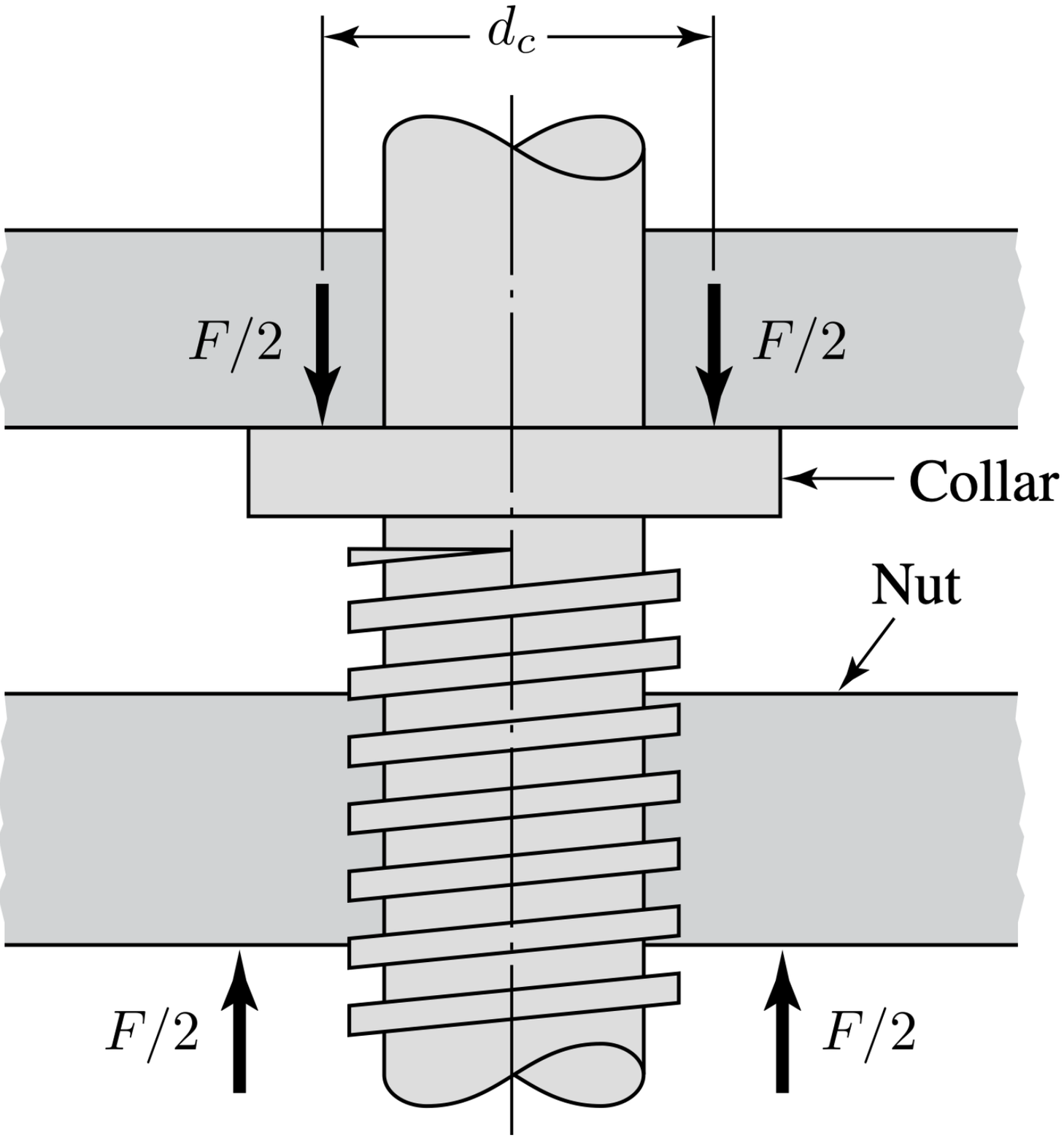

## Raising

$$T_R = \frac{F d_m}{2}\left(\frac{\pi d_m f \sec\alpha + \ell}{\pi d_m - \ell f \sec\alpha}\right)$$

## Lowering

$$T_L = \frac{F d_m}{2}\left(\frac{\pi d_m f \sec\alpha - \ell}{\pi d_m + \ell f \sec\alpha}\right)$$

# Collar Torque




$$T_C = F f_c d_c / 2$$

Added for both raising and lowering

# Efficiency

Torque required **without** friction over torque required **with** friction

$$e = \frac{P_{out}}{P_{in}} = \frac{T(f=0)}{T(f)}$$

$$e_R = \frac{F\ell}{2\pi(T_R + T_C)}$$

$$T_R = \frac{Fd_m}{2}\left(\frac{\cancel{\pi d_m f \sec\alpha} + \ell}{\pi d_m - \cancel{\ell f \sec\alpha}}\right)$$

$$T_L = \frac{Fd_m}{2}\left(\frac{\cancel{\pi d_m f \sec\alpha} - \ell}{\pi d_m + \cancel{\ell f \sec\alpha}}\right)$$

$$T_C = \cancel{F f_c d_c/2}$$

$$T(f=0) = \frac{F\ell}{2\pi}$$

# Example 1

What is the torque required to lower a 5 kN load using a square-thread power screw with 3 cm diameter, 6 mm pitch, 5 cm collar diameter, and friction coefficients $f = f_c = 0.1$?

$$T_R = \frac{F d_m}{2}\left(\frac{\pi d_m f \sec\alpha + \ell}{\pi d_m - \ell f \sec\alpha}\right)$$

$$T_L = \frac{F d_m}{2}\left(\frac{\pi d_m f \sec\alpha - \ell}{\pi d_m + \ell f \sec\alpha}\right)$$

$$T_C = F f_c d_c/2$$

$$e_R = \frac{F\ell}{2\pi(T_R + T_C)}$$

$F = 5{,}000$ N $\quad f = f_c = 0.1 \quad d_c = 0.05$ m

$d_m = d - p/2 = (0.03\text{ m} - (0.006\text{ m})/2) = 0.027$ m

$\ell = p = 0.006$ m $\quad \sec\alpha = 1$

$$T_L = \frac{(5{,}000\text{ N})(0.027\text{ m})}{2}\left(\frac{\pi(0.027\text{ m})(0.1)(1) - (0.006\text{ m})}{\pi(0.027\text{ m}) + (0.006\text{ m})(0.1)(1)}\right) \approx 1.96\text{ Nm}$$

$T_C = (5{,}000\text{ N})(0.1)(0.05\text{ m})/2 = 12.5$ Nm $\quad T_L + T_C \approx \boxed{14.46\text{ Nm}}$

# Example 2

A double-start acme-thread power screw with 50 mm diameter, 10 mm thread pitch, 100 mm collar diameter, and friction coefficients $f = f_c = 0.1$ is used to raise a 10 kN load. What is its efficiency?

$$T_R = \frac{F d_m}{2}\left(\frac{\pi d_m f \sec\alpha + \ell}{\pi d_m - \ell f \sec\alpha}\right)$$

$$T_L = \frac{F d_m}{2}\left(\frac{\pi d_m f \sec\alpha - \ell}{\pi d_m + \ell f \sec\alpha}\right)$$

$$T_C = F f_c d_c/2$$

$$e_R = \frac{F\ell}{2\pi(T_R + T_C)}$$

$F = 10{,}000$ N $\qquad f = f_c = 0.1 \qquad d_c = 0.1$ m

$d_m = d - p/2 = (50\text{ mm} - (10\text{ mm})/2) = 0.045$ m

$\ell = 2p = 2(10\text{ mm}) = 0.02$ m $\qquad \sec\alpha = \sec(14.5^\circ) \approx 1.033$

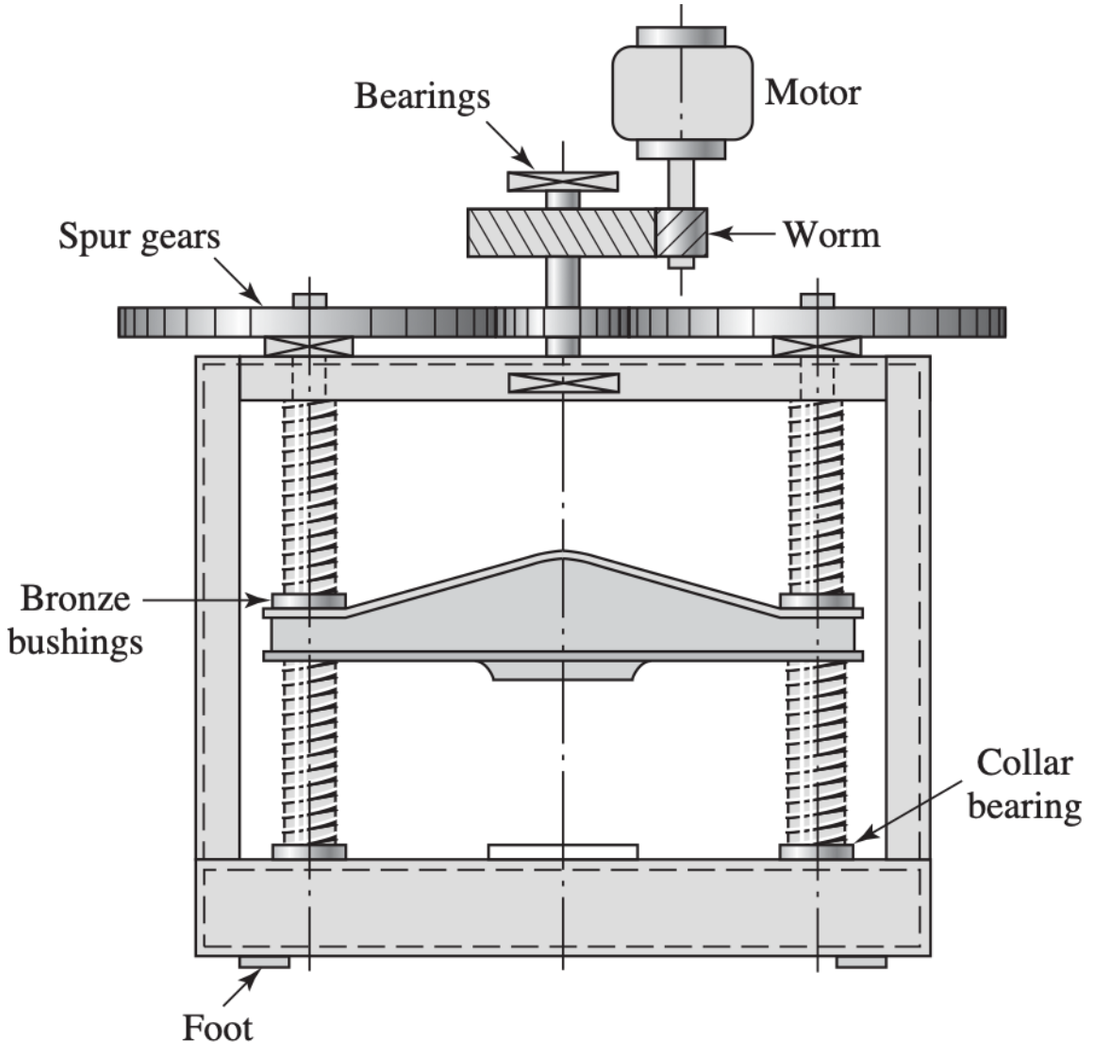
$$T_R = \frac{(10{,}000\text{ N})(0.045\text{ m})}{2}\left(\frac{\pi(0.045\text{ m})(0.1)(1.033) + (0.02\text{ m})}{\pi(0.045\text{ m}) - (0.02\text{ m})(0.1)(1.033)}\right) \approx 55.89\text{ Nm}$$

$$T_C = (10{,}000\text{ N})(0.1)(0.1\text{ m})/2 = 50\text{ Nm}$$

$$e_R \approx \frac{(10{,}000\text{ N})(0.02\text{ m})}{2\pi(55.89\text{ Nm} + 50\text{ Nm})} \approx 0.301 = \boxed{30.1\%}$$

# Example 3

The press shown conducts tension tests via two acme-thread screws with 2” diameter, 0.25” pitch, and coefficient of friction 0.05. The collar bearings have a collar diameter of 3.5” and coefficient of friction 0.08. The full-load motor speed is 1720 rev/min and the gears have a speed ratio of 60:1 with 95% efficiency. A slip clutch on the motor shaft prevents overloading.



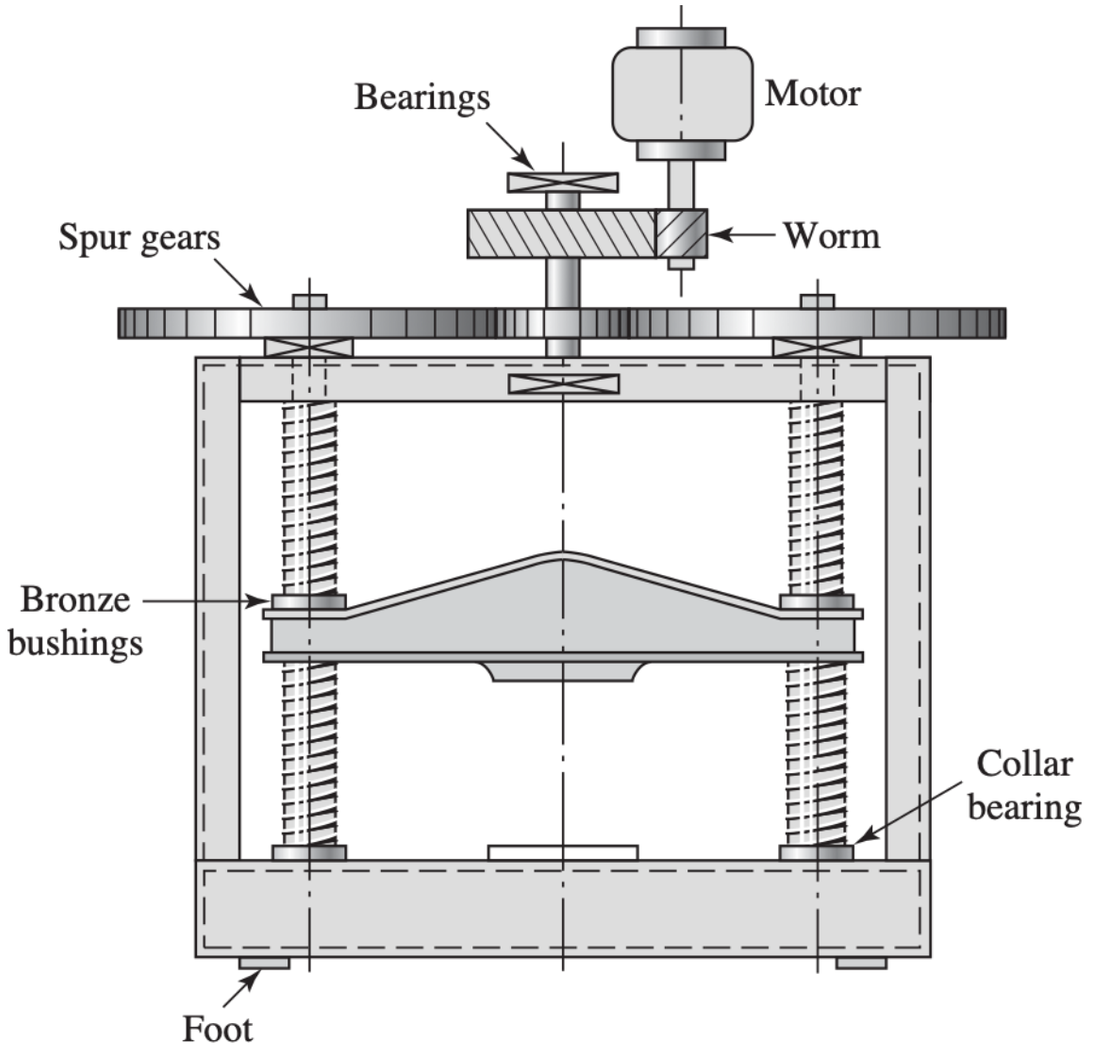

(a) How fast will the press head move?

$$\omega_m = \frac{1720 \text{ rev}}{1 \text{ min}} \qquad \frac{\omega_s}{\omega_m} = \frac{1}{60} \qquad v_s = \omega_s p$$

$$v_s = \left(\frac{1720 \text{ rev}}{1 \text{ min}}\right)\left(\frac{1}{60}\right)\left(\frac{0.25”}{1 \text{ rev}}\right) \approx \boxed{7.17 \text{ in/min}}$$

# Example 3

The press shown conducts tension tests via two acme-thread screws with 2” diameter, 0.25” pitch, and coefficient of friction 0.05. The collar bearings have a collar diameter of 3.5” and coefficient of friction 0.08. The full-load motor speed is 1720 rev/min and the gears have a speed ratio of 60:1 with 95% efficiency. A slip clutch on the motor shaft prevents overloading.




**(b) What should the horsepower rating of the motor be to achieve a max load of 5000 lbf?**

$$T_R = \frac{F d_m}{2}\left(\frac{\pi d_m f \sec\alpha + \ell}{\pi d_m - \ell f \sec\alpha}\right)$$

$$T_C = F f_c d_c/2 = (2500\text{ lbf})(0.08)(3.5”)/2 = 350\text{ lbf in}$$

$$d_m = (2” - 0.25”/2) = 1.875” \qquad \ell = p$$

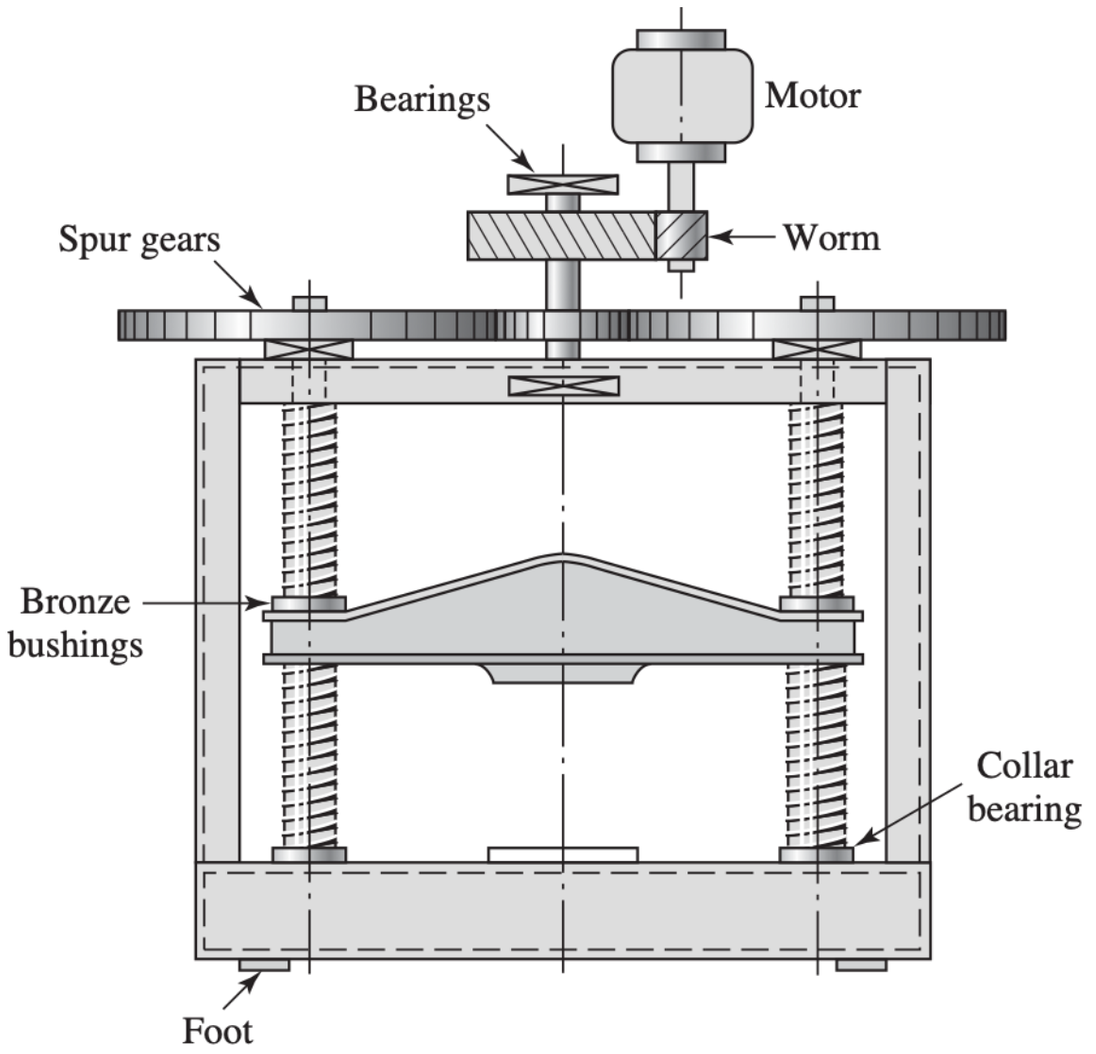
$$F = (5000\text{ lbf})/2 = 2500\text{ lbf} \qquad \sec\alpha = \sec(14.5^\circ) \approx 1.033$$

$$T_R = \frac{(2500\text{ lbf})(1.875”)}{2}\left(\frac{\pi(1.875”)(0.05)(1.033) + (0.25”)}{\pi(1.875”) - (0.25”)(0.05)(1.033)}\right) \approx 221.01\text{ lbf in}$$

$$T \approx 571.01\text{ lbf in}$$

# Example 3

The press shown conducts tension tests via two acme-thread screws with 2” diameter, 0.25” pitch, and coefficient of friction 0.05. The collar bearings have a collar diameter of 3.5” and coefficient of friction 0.08. The full-load motor speed is 1720 rev/min and the gears have a speed ratio of 60:1 with 95% efficiency. A slip clutch on the motor shaft prevents overloading.




(b) What should the horsepower rating of the motor be to achieve a max load of 5000 lbf?

$$e = 0.95 = \frac{P_{out}}{P_{in}} = \frac{2T\omega_{out}}{T_m\omega_m} = \frac{2T}{T_m}\frac{1}{60}$$

$$T_m = 2T/60/0.95 \approx 20.035 \text{ lbf in}$$

$$T \approx 571.01 \text{ lbf in}$$

$$P_m = T_m\omega_m = (20.035 \text{ lbf in})(1720 \text{ rev/min})$$

$$= (34{,}460 \text{ lbf in/min})\frac{1 \text{ hp}}{63{,}025 \text{ lbf in/min}} = \boxed{0.547 \text{ hp}}$$

# Practice

If a power screw is changed from square-threads to an acme-threads, its efficiency for raising vertical loads:

$$T_R = \frac{F d_m}{2}\left(\frac{\pi d_m f \sec\alpha + \ell}{\pi d_m - \ell f \sec\alpha}\right)$$

$$T_L = \frac{F d_m}{2}\left(\frac{\pi d_m f \sec\alpha - \ell}{\pi d_m + \ell f \sec\alpha}\right)$$

$$T_C = F f_c d_c/2$$

$$e_R = \frac{F\ell}{2\pi(T_R + T_C)}$$

A)

B)

C)

D)

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A) will stay the same

B) will increase

C) will decrease

D) may increase or decrease

# Practice

A single-start power screw is replaced by a double-start power screw with identical material, diameter, and pitch. The raising torque:

$$T_R = \frac{F d_m}{2}\left(\frac{\pi d_m f \sec\alpha + \ell}{\pi d_m - \ell f \sec\alpha}\right)$$

$$T_L = \frac{F d_m}{2}\left(\frac{\pi d_m f \sec\alpha - \ell}{\pi d_m + \ell f \sec\alpha}\right)$$

$$T_C = F f_c d_c/2$$

$$e_R = \frac{F\ell}{2\pi(T_R + T_C)}$$

A

B

C

D

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A) will stay the same

B) will increase

C) will decrease

D) may increase or decrease

# Practice

A 5 cm diameter power screw with 1 cm pitch is self-locking. Its coefficient of friction must be:

$$T_R = \frac{F d_m}{2}\left(\frac{\pi d_m f \sec\alpha + \ell}{\pi d_m - \ell f \sec\alpha}\right)$$

$$T_L = \frac{F d_m}{2}\left(\frac{\pi d_m f \sec\alpha - \ell}{\pi d_m + \ell f \sec\alpha}\right)$$

$$T_C = F f_c d_c/2$$

$$e_R = \frac{F\ell}{2\pi(T_R + T_C)}$$

A

B

C

D

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A) equal to 0.07

B) < 0.07

C) > 0.07

D) may be greater or less than 0.07

$$f = \frac{\ell}{\pi d_m}$$

# **ME2102:** Engineering Innovation and Modelling

## Power Screws

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